DHAMODHARAN S

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Aspiring AI practitioner with a keen interest in harnessing Machine Learning techniques to drive advancements in data-driven decision-making

EDUCATION

SRI KRISHNA COLLEGE OF TECHNOLOGY Coimbatore,Tamilnadu

Bachelor Of Engineering **Sep-2023-Present**

Major In Artificial Intelligence and Machine Learning
Cummulated GPA:8.14/10.0
Coursework in Computer Science,Machine Learning Algorithms ,Neural Networks

CHRIST THE KING MATRIC HR.SEC.SCHOOL Kumbakonam,Tamilnadu

HSC **Mar 2020-Apr2023**

Major in Mathematics with Biology
Percentage of Marks:91%
Coursework in Physics, Biology, Chemistry, Maths

SKILLS

- **Programming Languages:** PYTHON,C++,JAVA
- **Machine Learning Algorithms, Deep Neural Networks-Convolution Neural Network**
- **Data Structures and Algorithms**
- **Database Management System:** SQL
- Front End Development(HTML,CSS,JAVASCRIPT,REACt
- **Frameworks & Libraries:** TensorFlow , Django, Flask, Streamlit, Scikit-Learn, Numpy, Pandas, Matplotlib, Seaborn
- **Programming Tools:** Anaconda, Google Colab, VS code,Aurdino IDE
- **Microcontrollers:** RasberryPi 3b+, AurdinoUNO, Aurdino Nano, STM32(bluepill), ESP-32
- **Version Control:** Git and Gitub

AREAS OF INTERESTS

- Machine Learning
- Generative AI
- Deep Learning
- Computer Vision
- Natural Language Processing
- Internet Of Things

LANGUAGES

- **Tamil**-Native
- **English-**Proficient

PROJECTS

1.AI-Powered Text Summarizer and Speech Converter

Designed and implemented a web application using Flask that extracts text from uploaded or scanned documents via **PyTesseract**, summarizes the content, and converts it into speech using the Pre-trained **Coqui TTS** model. Integrated **OpenCV** for image processing, providing a user-friendly solution to simplify document reading.

Technologies Used: Python, Flask, Coqui TTS, PyTesseract, OpenCV

2.Speech to ISL Translator using Speech Recognition & NLP Techniques:

An AI based tool to translate the given Speech to Indian Sign Language Gesture Videos using NLP Techniques.The Process involves Speech Recognition,Text Processing & fetching Video of corresponding sign language gestures.

Technologies Used: Python, Google Translate API, Natural Language Toolkit Library

3.A Web based Multilple Disease Prediction System using Machine Learning:

An Web App based tool that utilizes Machine Learning Classification algorithms (Random Forest,Decision Trees,SVM) to predict the presence of diseases such as diabetes, heart failure, lung cancer, and obesity based on user-inputted parameters. The system incorporates data preprocessing and model training to ensure accurate predictions

Technologies Used: Python, Scikit-Learn, Streamlit

4.Deep Learning Based Skin Cancer Classification System:

A Web Interface integrated with Deep learning Technique Convolutional neural networks (CNN) to classify skin lesion images as benign or malignant. The model leverages advanced image processing techniques for accurate diagnosis, aiding in early detection of skin cancer.

Technologies Used: Python, TensorFlow, Keras, Streamlit

5.Line Follower bot using Arduino and IoT Sensors:

Designed an autonomous robot that follows a black line using infrared (IR) sensors for detection. The bot utilizes Arduino for control and processing, enabling real-time response to sensor input.

Technologies Used: Embedded C **,** Arduino UNO, Arduino IDE

ACHIEVEMENTS

1.Elite Silver + Topper Certificate On NPTEL (Deep Learning by IIT Ropar) **Dec-2024**

Earned an Elite+Silver certificate, ranking in the **Top 5%** of the NPTEL course "**Deep Learning by IIT Ropar**" with 75%. Gained expertise in **Neural Networks**, including architectures like CNN, RNN, LSTM, and GRU, with a solid understanding of backpropagation, attention mechanisms, and encoder-decoder models

2.Elite Certificate On NPTEL (Introduction to Machine Learning) **Mar-2024**

Earned an **Elite** certificate with **68**% in the NPTEL course "**Introduction to Machine Learning**." Gained a strong foundation in core ML concepts, algorithms, and practical applications through comprehensive coursework and assessments.

3.Runner Up in National Level Technical Event Organized by PSG Tech-SriShti 2K23 **Oct-2023**

Secured Runner-Up position in a national-level electronics-based event, Techilluminate, organized by PSG Tech-SriShti 2K23. Demonstrated technical expertise in innovative electronics solutions and practical problem-solving

CERTIFICATIONS

1.Deep Learning by IIT Ropar with 75% -*NPTEL* **Dec-2024**
2.Scikit-Learn For Machine Learning Classification Problems by Coursera Project Network-*Coursera* **Apr-2024**
3.Machine Learning with Python by IBM- *Coursera* **Mar-2024**
4.Python for Data Analysis-Pandas & Numpy by Coursera project network- *Coursera* **Mar-2024**
5.Introduction to Machine Learning with 68 % - *NPTEL* **Mar-2024**
6.Data Science Using Python by IBM - *Coursera* **Feb-2024**

PARTICIPATIONS

1. Smart India Hackathon-2024 - Participated in Internal Hackathon for SIH-2024 and Proposed the Solution - **An AI based Tool for Translating Indian Sign Language gestures from a Recognized Speech.**

2. MSME Idea Hackathon - Participated in Idea Proposal Hackathon MSME-2024 and Proposed the Solution - **Precision Human Detection in CSSR Operations : AI-Driven Drone Technology Integrated Thermal Imaging and IR-UWB Radar.**

3.Shastra Smart City Challenge IIT Madras- Participated in Idea Pitch Hackathon Organized by IIT Madras and Proposed the Solution - **GLOW-GREEN-An Innovative Approach to Capture the Atmospheric C02 by Direct Air Capture Methodology**